

Toxic Habit Named Entity Recognition in Spanish Clinical Texts

A Character-Aware BiLSTM-CRF Baseline for BioCreative IX ToxHabits

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Abstract

This report presents an end-to-end named entity recognition (NER) system for extracting toxic-habit mentions from Spanish clinical case reports. The original BRAT span annotations were converted to token-level BIO sequences while preserving document identifiers and global character offsets. Long documents were segmented with sentence-aware greedy chunking and reconstructed at document level for evaluation. The proposed model combines learned word embeddings, a character-level convolutional neural network, a bidirectional long short-term memory encoder, and a linear-chain conditional random field. On an internal document-level held-out test set, the model obtained strict entity-level precision of 0.7600, recall of 0.8296, micro-F1 of 0.7933, and macro-F1 of 0.8454. Performance was highest for *Cannabis* (F1 0.9635) and lowest for the heterogeneous *Drug* category (F1 0.7541). Error analysis shows that the dominant limitations are exact span boundary selection, context-dependent annotation decisions, and generalization to unseen entity surface forms rather than confusion between entity types. The system therefore provides a strong, interpretable classical neural baseline and a clear reference point for future comparison with pretrained Spanish biomedical language models.

1 Introduction

Substance use and abuse are clinically relevant factors that influence disease risk, treatment decisions, and medication effects. Much of this information is recorded in free-text clinical narratives rather than in structured fields. BioCreative IX introduced the ToxHabits track to study automatic extraction of tobacco, alcohol, cannabis, and other drug-related information from Spanish clinical text [?, ?].

This project addresses the ToxNER mention-recognition subtask as a sequence-labeling problem. Its main objectives are:

- to build a reliable BRAT-to-BIO preprocessing pipeline with exact character-offset preservation;
- to implement a character-aware neural sequence tagger and a linear-chain CRF from modular PyTorch components;
- to evaluate predictions at strict entity level after document reconstruction; and
- to identify whether the remaining errors arise from entity detection, label classification, span boundaries, or lexical generalization.

The selected architecture follows the classical CNN-BiLSTM-CRF family introduced for end-to-end sequence labeling [?] and related character-aware neural NER systems [?]. Unlike a transformer baseline, it makes the contribution of morphology, contextual encoding, and structured decoding explicit.

2 Dataset and exploratory analysis

2.1 Corpus

ToxHabits-NER is a gold-standard corpus of Spanish clinical case reports released for the BioCreative IX ToxHabits shared task [?]. The complete challenge corpus contains 1,499 documents. At project development time, 1,199 documents had accessible gold annotations and were used to construct internal training, validation, and held-out test partitions; the 300-document official test portion was not used for the results reported here.

Annotations are represented in BRAT standoff format and assign each mention to one of four semantic classes:

Class	Interpretation
Drug	illicit drugs, medications, toxic substances, and related expressions
Alcohol	alcoholic beverages and alcohol-consumption expressions
Tobacco	tobacco products, smoking, and related expressions
Cannabis	cannabis products and cannabis-consumption expressions

Thirty-one documents contained no annotated entities and were retained as negative samples. This is important because removing them would inflate entity density and under-represent documents in which the correct prediction is entirely outside the entity classes.

2.2 Entity and document statistics

The corpus is imbalanced: **Drug** is the dominant label by a substantial margin, while the other three classes have fewer annotated mentions. A total of 730 gold entities contain multiple tokens; the remaining mentions are single-token entities. Multi-token spans include chemically specific names, coordinated expressions, parenthetical aliases, and clinically contextual phrases.

Documents are too long to process efficiently as single recurrent sequences. The mean document length is 617.5 tokens and 3,458.2 characters. Token-length percentiles are shown in Table 1.

Table 1: Document and sentence length statistics after Spanish tokenization.

Unit	Mean	75th percentile	95th percentile	99th percentile
Document, tokens	617.5	808.0	1,526.1	2,181.5
Sentence, tokens	—	—	55.0	87.0

Maximum observed sentence length: 317 tokens. Mean and 75th percentile were not retained for the sentence-level analysis.

Initial sentence segmentation produced one entity crossing a predicted boundary: *comps. de Metasedin*. Inspection showed that the split was caused by treating the abbreviation *comps.* as a sentence ending. A custom boundary rule corrected this case before chunk generation.

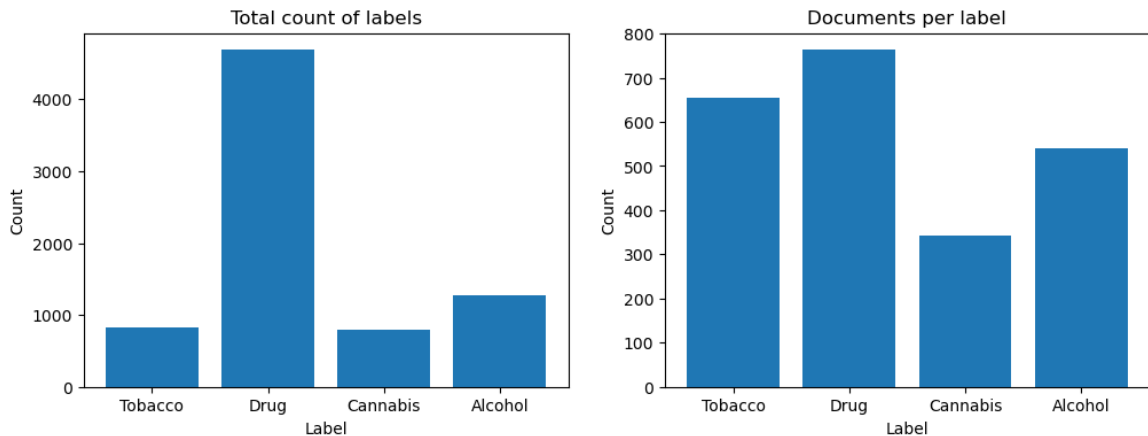


Figure 1: Corpus imbalance at entity and document level. Drug mentions dominate the annotation inventory.

3 Preprocessing pipeline

3.1 BRAT-to-BIO conversion

Each BRAT annotation is a tuple

$$a_k = (\ell_k, s_k, e_k, u_k),$$

where ℓ_k is the entity class, $[s_k, e_k)$ is the half-open character interval in the original document, and u_k is the annotated text. Tokenization produces tokens

$$w_i = (v_i, b_i, d_i),$$

with surface form v_i and character interval $[b_i, d_i)$. A token overlaps an entity whenever

$$b_i < e_k \quad \wedge \quad d_i > s_k.$$

The first overlapping token receives B- ℓ_k , subsequent overlapping tokens receive I- ℓ_k , and all other tokens receive O. Conversion was validated by comparing every annotated span against the covered token interval; no entity boundaries were lost.

The BIO label set contains nine tags:

$$\mathcal{Y} = \{O\} \cup \{B-c, I-c : c \in \{\text{Drug, Alcohol, Tobacco, Cannabis}\}\}.$$

3.2 Document-level splitting and chunking

Documents were assigned to training, validation, and test partitions *before* chunking. This prevents chunks originating from the same clinical report from appearing in multiple partitions. The split was balanced with respect to document-level label presence.

Within each partition, consecutive sentences were greedily packed into chunks with a target maximum of 256 tokens. An overlong individual sentence was kept as a single unit rather than silently dropping tokens or splitting an entity. Every sample retained:

- the original document identifier;
- the chunk identifier and order;
- token strings and token-level character offsets; and
- the corresponding BIO sequence.

At inference time, chunk predictions are mapped back to global character intervals and merged by document identifier. The identifier is metadata, not a learned model input. This separation allows minibatch training while preserving exact document-level provenance.

3.3 Vocabulary and batching

The word and character vocabularies were constructed from the training partition only. Unseen items in validation and test data map to <unk>; padding uses dedicated <pad> symbols. Character sequences are padded within each batch to the longest token, while token sequences are padded to the longest chunk. A Boolean token mask prevents padded positions from contributing to the CRF objective or decoding.

4 Model

4.1 Character-level CNN

For token w_i with characters $(c_{i,1}, \dots, c_{i,m_i})$, the character embedding layer produces

$$C_i = [e_{i,1}^c; \dots; e_{i,m_i}^c] \in \mathbb{R}^{m_i \times d_c}.$$

For a convolutional kernel of width k , local features are

$$\mathbf{q}_{i,j}^{(k)} = \text{ReLU}\left(\mathbf{W}^{(k)} \text{vec}(C_{i,j:j+k-1}) + \mathbf{b}^{(k)}\right),$$

followed by max-over-time pooling:

$$\mathbf{g}_i^{(k)} = \max_j \mathbf{q}_{i,j}^{(k)}.$$

The final character representation concatenates outputs for kernel widths 3, 4, and 5:

$$\mathbf{g}_i = [\mathbf{g}_i^{(3)}; \mathbf{g}_i^{(4)}; \mathbf{g}_i^{(5)}].$$

This representation captures subword patterns useful for chemical names, inflection, punctuation, prefixes, and previously unseen spellings.

4.2 Word representation and BiLSTM encoder

The model concatenates a learned word embedding e_i^w with the character representation:

$$\mathbf{x}_i = [e_i^w; \mathbf{g}_i].$$

A bidirectional LSTM computes left-to-right and right-to-left contextual states:

$$\begin{aligned} \vec{\mathbf{h}}_i &= \text{LSTM}_{\rightarrow}(\mathbf{x}_i, \vec{\mathbf{h}}_{i-1}), & \overleftarrow{\mathbf{h}}_i &= \text{LSTM}_{\leftarrow}(\mathbf{x}_i, \overleftarrow{\mathbf{h}}_{i+1}), \\ \mathbf{h}_i &= [\vec{\mathbf{h}}_i; \overleftarrow{\mathbf{h}}_i]. \end{aligned}$$

A linear classifier maps $\mathbf{h}_i$ to emission scores $\mathbf{z}_i \in \mathbb{R}^{|\mathcal{Y}|}$:

$$\mathbf{z}_i = \mathbf{W}_o \mathbf{h}_i + \mathbf{b}_o.$$

4.3 Linear-chain CRF

Independent token classification cannot directly represent dependencies such as the requirement that an **I-Drug** tag should follow a compatible beginning or inside tag. The linear-chain CRF assigns a global score to an entire tag sequence $\mathbf{y}$:

$$S(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^T z_{i, y_i} + \sum_{i=1}^T A_{y_{i-1}, y_i},$$

where $A_{a,b}$ is a learned transition score, including start and end transitions. The conditional probability is

$$p(\mathbf{y} \mid \mathbf{x}) = \frac{\exp S(\mathbf{x}, \mathbf{y})}{\sum_{\mathbf{y}' \in \mathcal{Y}^T} \exp S(\mathbf{x}, \mathbf{y}')}.$$

Training minimizes the negative log-likelihood

$$\mathcal{L}(\mathbf{x}, \mathbf{y}^*) = -S(\mathbf{x}, \mathbf{y}^*) + \log \sum_{\mathbf{y}' \in \mathcal{Y}^T} \exp S(\mathbf{x}, \mathbf{y}'),$$

where the partition function is computed with the forward dynamic program. At inference time, Viterbi decoding returns

$$\hat{\mathbf{y}} = \arg \max_{\mathbf{y} \in \mathcal{Y}^T} S(\mathbf{x}, \mathbf{y}).$$

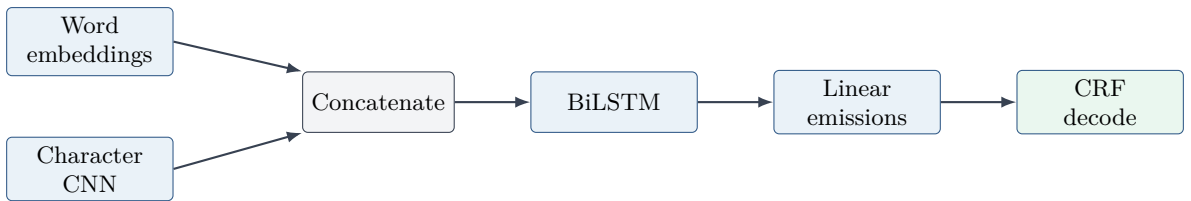


Figure 2: Model architecture. Word-level and character-level features are combined before contextual and structured sequence modeling.

4.4 Configuration

5 Evaluation protocol

Predictions are decoded per chunk, restored to global document offsets, and evaluated as labeled spans. A prediction counts as a true positive only when both its semantic class and character boundaries exactly match a gold entity. Consequently, partial overlap is counted as one false positive and one false negative.

For a class c ,

$$P_c = \frac{TP_c}{TP_c + FP_c}, \quad R_c = \frac{TP_c}{TP_c + FN_c}, \quad F_{1,c} = \frac{2P_c R_c}{P_c + R_c}.$$

Table 2: Model configuration used in the reported implementation.

Component	Value
Word embedding dimension	100
Character embedding dimension	30
Character convolution widths	{3, 4, 5}
Filters per convolution width	30
Concatenated character dimension	90
BiLSTM hidden dimension, per direction	128
BiLSTM layers	1
Dropout	0.2
Output tags	9
Chunk target length	256 tokens
Optimization	AdamW

Micro-F1 aggregates TP , FP , and FN before computing the score:

$$F_1^{\text{micro}} = \frac{2 \sum_c TP_c}{2 \sum_c TP_c + \sum_c FP_c + \sum_c FN_c},$$

whereas macro-F1 weights the four classes equally:

$$F_1^{\text{macro}} = \frac{1}{4} \sum_{c=1}^4 F_{1,c}.$$

Token accuracy is reported only as a diagnostic because the dominant 0 tag makes it overly optimistic. The invalid BIO rate measures the proportion of predicted sequences containing at least one structurally invalid transition according to the BIO scheme.

Diagnostic evaluation additionally separates single-token from multi-token entities and previously observed from unseen surface forms. Here, *seen* means that the complete normalized entity string appeared in the training partition; therefore the appropriate term is *unseen surface form*, not strictly an out-of-vocabulary token.

6 Results

6.1 Overall performance

Table 3 reports strict entity-level performance. Aggregate scores were recomputed directly from the class-level $TP/FP/FN$ counts produced by the final evaluator.

Table 3: Strict entity-level results. The test partition is an internal held-out subset of the annotated development corpus.

Split	Precision	Recall	Micro-F1	Macro-F1	TP	FP / FN
Validation	0.7411	0.8111	0.7746	0.8192	627	219 / 146
Internal test	0.7600	0.8296	0.7933	0.8454	589	186 / 121

The test-set recall exceeds precision by approximately seven points, indicating a moderately recall-oriented model. The near-0.994 token accuracy is not inconsistent with the lower entity F1: most tokens belong to the 0 class, while a single incorrect boundary token invalidates an otherwise overlapping entity under strict matching.

6.2 Performance by entity type

Cannabis is recognized most reliably, followed by **Tobacco**. The **Drug** class is considerably harder. It has the largest support, the broadest semantic scope, and the most varied surface forms, ranging from chemical

Table 4: Per-class strict entity-level performance. Gold support equals $TP + FN$.

Split	Class	Precision	Recall	F1	TP	FP	FN
Validation	Drug	0.6949	0.7842	0.7368	378	166	104
	Alcohol	0.7868	0.8295	0.8075	107	29	22
	Tobacco	0.7857	0.8148	0.8000	66	18	15
	Cannabis	0.9268	0.9383	0.9325	76	6	5
Internal test	Drug	0.7280	0.7820	0.7541	348	130	97
	Alcohol	0.7333	0.8871	0.8029	110	40	14
	Tobacco	0.8442	0.8784	0.8609	65	12	9
	Cannabis	0.9429	0.9851	0.9635	66	4	1

compounds and commercial names to generic descriptions and consumption-related expressions. Its lower precision accounts for most false-positive mass.

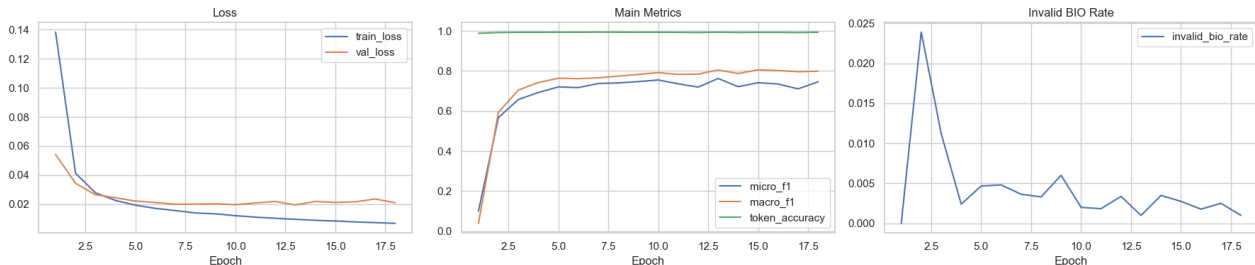
6.3 Span length and lexical novelty

Table 5: Diagnostic F1 by span length and training-set surface-form status. Values with very small support may be unstable and are interpreted descriptively.

Split	Class	One-token F1	Multi-token F1	Seen F1	Unseen-form F1
Validation	Drug	0.7444	0.6866	0.8035	0.1524
	Alcohol	0.8061	1.0000	0.8249	0.2500
	Tobacco	0.8049	0.0000	0.7975	1.0000
	Cannabis	0.9325	0.0000	0.9500	0.0000
Internal test	Drug	0.7640	0.6733	0.8128	0.3243
	Alcohol	0.8000	0.8889	0.8059	0.0000
	Tobacco	0.8667	0.0000	0.8725	0.0000
	Cannabis	0.9635	0.0000	0.9635	0.0000

The most meaningful pattern is the large seen-unseen gap for **Drug**: 0.8035 versus 0.1524 on validation and 0.8128 versus 0.3243 on test. The model therefore relies strongly on previously observed lexical forms. Character features improve morphological sensitivity but cannot fully infer the semantics of a novel substance name from spelling alone. Values of exactly zero or one for rare class-subset combinations should not be treated as robust estimates without corresponding support counts.

6.4 Training dynamics

**Figure 3:** Training history. Validation loss and entity-level F1 improve rapidly during the first epochs and subsequently plateau while training loss continues to decline, indicating mild overfitting. Token accuracy remains close to 0.99 throughout and is therefore not a useful checkpoint-selection metric.

Most generalization gains occur within approximately the first 7–10 epochs. Later epochs continue to reduce training loss without a stable improvement in validation F1. Model selection should consequently use validation entity-level micro-F1 and early stopping rather than training loss or token accuracy.

7 Error analysis

7.1 Boundary errors

Cross-label confusion is rare: the recorded error analysis contained only one overlapping entity assigned a different class (gold `Alcohol`, predicted `Drug`: *coca-cola*). The dominant failure mode is selecting the wrong extent for an otherwise semantically related span.

Table 6: Representative boundary-error patterns.

Pattern	Gold span	Predicted span
Alias only	<i>fenciclidina (PCP)</i>	<i>PCP</i>
Alias only	<i>dibutilona (bk-DMBDB)</i>	<i>bk-DMBDB</i>
Dropped modifier	<i>clorhidrato de cocaína</i>	<i>cocaína</i>
Dropped modifier	<i>muchos tóxicos</i>	<i>tóxicos</i>
Over-extension	<i>heroína</i>	<i>heroína-fumada</i>
Over-extension	<i>opiáceos</i>	<i>opiáceos ilegales</i>
Coordination	<i>MDMA</i>	<i>MDMA o “Molly”</i>
Negation scope	<i>no consumidora</i>	<i>consumidora</i>
Quantifier scope	<i>ningún tipo de tóxico</i>	<i>tóxico</i>
Class and extent	<i>bebidas energéticas (Drug)</i>	<i>bebidas (Alcohol)</i>

These errors form four recurring groups:

- **Parenthetical aliases:** the model often selects either the full form or the abbreviation rather than the complete gold construction.
- **Modifiers and heads:** generic heads such as *tóxicos* or specific substance names are detected while left or right modifiers are omitted or added.
- **Coordination:** conjunctions and neighboring aliases encourage the CRF to extend a span across adjacent mentions.
- **Negation and quantification:** the lexical trigger is recognized, but the annotation scheme sometimes includes a preceding negative or quantifying phrase that the model excludes.

The CRF captures local tag dependencies but does not by itself learn the semantic policy that determines whether a modifier, alias, or negation belongs inside a gold span. Strict entity scoring magnifies this limitation because even a one-token disagreement invalidates the full prediction.

7.2 Frequent false positives and false negatives

Table 7: Most frequent false-positive and false-negative surface forms.

False positives			False negatives		
Class	Surface form	Count	Class	Surface form	Count
Alcohol	<i>alcohol</i>	17	Drug	<i>abstinencia</i>	10
Drug	<i>consumo</i>	9	Drug	<i>buprenorfina / naloxona</i>	4
Drug	<i>benzodiacepinas</i>	6	Drug	<i>droga</i>	4
Tobacco	<i>fumador</i>	6	Tobacco	<i>cigarrillos</i>	4
Drug	<i>metanfetamina</i>	5	Alcohol	<i>beber</i>	3
Alcohol	<i>beber</i>	4	Drug	<i>tóxicos</i>	3
Drug	<i>desintoxicación</i>	4	Drug	<i>consumir</i>	2
Drug	<i>metadona</i>	4	Drug	<i>colas</i>	2
Drug	<i>buprenorfina / naloxona</i>	4	Drug	<i>consumo</i>	2
Drug	<i>heroína</i>	3	Alcohol	<i>combinados</i>	2

Several forms occur in both lists, including *beber*, *consumo*, *tóxicos*, and *buprenorfina / naloxona*. This suggests that the model has learned their lexical association with toxic habits but does not consistently reproduce

the context-dependent annotation decision. A surface-form list alone cannot distinguish model error from contextual variation: mentions may occur in negated statements, treatment descriptions, family history, or another scope that the annotation guidelines exclude.

7.3 Interpretation

The error profile supports three conclusions:

1. **Entity typing is not the primary bottleneck.** Once an overlapping entity is found, its semantic class is almost always correct.
2. **Exact span policy is difficult.** Parentheses, aliases, modifiers, coordination, and negation account for many strict-match errors.
3. **Lexical memorization remains substantial.** The large unseen-surface-form gap, particularly for **Drug**, reveals limited semantic generalization despite character-level modeling.

8 Limitations and future work

The reported results should be interpreted with the following limitations:

- Evaluation uses one internal split of the public annotated corpus rather than the official challenge test set.
- The reported experiment represents a single training run; variation across random seeds was not measured.
- Diagnostic multi-token and unseen-form F1 values were retained without subset support counts, making extreme values unstable.
- Strict matching does not distinguish a nearly correct boundary from a completely unrelated prediction.
- Learned word embeddings provide no external distributional knowledge, limiting recognition of unseen biomedical terms.
- Some apparent errors may reflect context-sensitive or ambiguous annotation guidelines rather than simple lexical mistakes.

A minimal extension of the evaluator should report both strict and overlap-based F1, classify boundary-only errors into under- and over-extension, and attach support counts to every diagnostic subset. The most informative modeling extension would be a comparison against a pretrained Spanish or Spanish-biomedical transformer with a token-classification head, optionally followed by a CRF. Such a model could improve contextual scope and unseen-form generalization while preserving the current preprocessing and document-reconstruction pipeline.

9 Conclusion

This project implemented a complete clinical NER workflow from raw BRAT annotations to document-level entity predictions. The CharCNN-BiLSTM-CRF system achieved a strict entity-level micro-F1 of 0.7933 and macro-F1 of 0.8454 on an internal held-out test set. It performed particularly well for Cannabis and Tobacco mentions, while the diverse Drug category remained the main source of errors.

The detailed analysis reveals a more informative result than the aggregate score alone: the model generally distinguishes entity types correctly, but exact boundary selection and unseen surface forms remain challenging. The project therefore establishes a transparent classical neural baseline and demonstrates the importance of offset-preserving preprocessing, leakage-safe document splitting, structured decoding, and entity-level error analysis in biomedical NLP.

References

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A Reproducibility checklist

- Documents are split before sentence segmentation and chunking.
- Word and character vocabularies are fitted on training data only.
- BRAT offsets are checked against tokenized spans after BIO conversion.
- Padding positions are masked in the CRF loss and Viterbi decoder.
- Predictions are reconstructed using document identifiers and global offsets.
- Checkpoint selection uses validation entity-level F1 only.
- The internal test set is evaluated after checkpoint selection.
- Aggregate metrics are recomputed from the same per-class $TP/FP/FN$ output.